

useEffect:

The `useEffect` Hook in React allows you to perform side effects in functional components, such as data fetching, subscriptions, or manual DOM manipulation. It runs after the component renders and provides a way to synchronize your component with external systems.

```
1 import { useState, useEffect } from "react";
2
3 function Basic() {
4   const [count, setCount] = useState(0);
5
6   useEffect(() => {
7     setTimeout(() => {
8       setCount((count) => count + 1);
9     }, 1000);
10  });
11
12  return <h1>I've rendered {count} times!</h1>;
13 }
14
15 export default Basic;
16
```

```
1 import { useState, useEffect } from "react";
2
3 function OnlyOnce() {
4   const [count, setCount] = useState(0);
5
6   useEffect(() => {
7     setTimeout(() => {
8       setCount((count) => count + 1);
9     }, 1000);
10  }, []);
11
12  return <h1>I've rendered {count} times!</h1>;
13 }
14
15 export default OnlyOnce;
16
```

```

1  import { useState, useEffect } from "react";
2
3  function Dependency() {
4    const [count, setCount] = useState(0);
5    const [calculation, setCalculation] = useState(0);
6
7    useEffect(() => {
8      setCalculation(() => count * 2);
9    }, [count]); // <- add the count variable here
10
11   const handleClick = () => {
12     setCount((c) => c + 1);
13   };
14
15   return (
16     <>
17       <h1>Count: {count}</h1>
18       <button onClick={handleClick}>Button</button>
19       <h1>Multiple of count by 2: {calculation}</h1>
20     </>
21   );
22 }
23
24 export default Dependency;
25

```

API:

Get example:

```

1  import React, { useState, useEffect } from "react";
2
3  const GetApiData = () => {
4      const [items, setItems] = useState([]);
5      const [dataIsLoaded, setDataIsLoaded] = useState(false);
6      const [error, setError] = useState(null);
7
8      useEffect(() => {
9          const fetchData = async () => {
10             try {
11                 const response = await fetch(
12                     "https://jsonplaceholder.typicode.com/users"
13                 );
14                 const result = await response.json();
15                 console.log("i have a result", result);
16                 setItems(result);
17                 setDataIsLoaded(true);
18             } catch (error) {
19                 setError(error);
20             }
21         };
22
23         fetchData();
24     }, []);
25

```

```

26     // if data is not loaded
27
28     if (!dataIsLoaded) {
29         return (
30             <div>
31                 <h1>Please wait some time....</h1>
32             </div>
33         );
34     }
35
36     return (
37         <div>
38             <h3>Fetch data from an API in React</h3>
39             <div>
40                 {items.map((item) => (
41                     <div key={item.id}>
42                         <ol>
43                             <div>
44                                 <strong>User Name: {item.username}</strong>
45                             </div>
46                             <div>Full Name: {item.name}</div>
47                             <div>User Email: {item.email}</div>
48                         </ol>
49                     </div>
50                 ))}
51             </div>
52         </div>
53     );
54 };
55 export default GetApiData;

```

Post API Code:

```
1  import React, { useState } from "react";
2
3  function PostDataApi() {
4    const [title, setTitle] = useState("");
5    const [body, setBody] = useState("");
6    const [responseMessage, setResponseMessage] = useState("");
7
8    const handleSubmit = async (event) => {
9      event.preventDefault(); // Prevent default form submission behavior
10
11      try {
12        const response = await fetch(
13          "https://jsonplaceholder.typicode.com/posts",
14          {
15            method: "POST",
16            headers: {
17              "Content-Type": "application/json",
18            },
19            body: JSON.stringify({ title, body, userId: 1 }), // Send data as JSON string
20          }
21        );
22
23        const data = await response.json();
24        setResponseMessage(`Post created successfully! ID: ${data.id}`);
25        setTitle("");
26        setBody("");
27      } catch (error) {
28        setResponseMessage(`Error: ${error.message}`);
29      }
30    };
31  }
```

```

31
32     return (
33         <div>
34             <h2>Posting data to verify post call</h2>
35             <form onSubmit={handleSubmit}>
36                 <div>
37                     <label htmlFor="title">Title:</label>
38                     <input
39                         type="text"
40                         id="title"
41                         value={title}
42                         onChange={(e) => setTitle(e.target.value)}
43                     />
44                 </div>
45                 <div>
46                     <label htmlFor="body">Body:</label>
47                     <textarea
48                         id="body"
49                         value={body}
50                         onChange={(e) => setBody(e.target.value)}
51                     ></textarea>
52                 </div>
53                 <button type="submit">Submit Post</button>
54             </form>
55             {responseMessage && <p>{responseMessage}</p>}
56         </div>
57     );
58 }
59
60 export default PostDataApi;
61

```